

ADHDNet: A Lightweight Adaptive Multimodal 3D CNN for ADHD Classification Using Structural MRI and Phenotypic Features

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Abstract. ADHD is diagnosed mainly through behavioral and clinical assessment, which has led to interest in neuroimaging-based methods. Deep learning models for structural MRI, however, often use large 3D networks, and multimodal models add further cost through fusion components. This study examines whether similar predictive performance can be obtained with a much smaller multimodal 3D model. For this purpose, ADHDNet is proposed: a lightweight 3D CNN that combines depthwise-separable convolutions and efficient channel attention with a gated fusion of MRI and phenotypic features. On 497 ADHD-200 subjects from five sites, an accuracy of 90.00% and an AUC of 0.9543 were obtained by ADHDNet (71,552 parameters; 0.19 GFLOPs per volume) on a held-out validation set, and $87.52 \pm 4.61\%$ accuracy (AUC 0.951 ± 0.017) was obtained in 5-fold cross-validation. Compared with a conventional Conv3D multimodal model, 94.8% fewer parameters and 97.2% fewer FLOPs were required at a similar cross-validated AUC. However, close-to-chance performance was observed for the MRI-only variant in an ablation study. The results show that similar multimodal performance can be obtained with a much smaller 3D model, but also that this performance comes mainly from the phenotypic features, so the additional contribution of structural MRI remains unclear.

Keywords: ADHD · Structural MRI · Multimodal learning · 3D CNN · Lightweight deep learning

1 Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder marked by persistent inattention, hyperactivity, and impulsivity. It is diagnosed mainly through behavioral and clinical assessment, which can be difficult because its symptoms overlap with those of other developmental and psychiatric conditions. This has motivated work on neuroimaging-based methods that look for brain patterns associated with ADHD [1]–[3].

Structural magnetic resonance imaging (sMRI) shows brain anatomy without radiation and has been widely used for this purpose [4]. In early studies, conventional classifiers were applied to measures such as gray-matter volume

and cortical thickness [5]. Deep learning later made it possible to learn features directly from the images, and recent ADHD studies have used convolutional, multimodal, attention-based, and ensemble models [6]–[9]. Multimodal models also add phenotypic features such as age, sex, and clinical ratings. As these models grow deeper and wider and use more complex fusion, the computational and memory cost rises. This matters when computing resources are limited, and compact models, such as the few-thousand-parameter network of Deepika et al. [10], have been explored for this reason.

Three questions follow from this. Can structural MRI and phenotypic features be modeled together at a much lower computational cost? Can a small 3D model keep the performance of a larger one? And does the resulting performance come from the MRI, from the phenotypic features, or from particular model components? No new building blocks are proposed. Instead, established efficient components are combined in a model called *ADHDNet*, and this combination is tested under controlled conditions. The main contributions of this study are:

1. A **lightweight multimodal 3D model** that learns from MRI and phenotypic features together, with far fewer parameters and FLOPs than a conventional Conv3D multimodal model.
2. A **lightweight MRI encoder** that combines depthwise-separable 3D convolutions with efficient channel attention to keep 3D feature extraction while reducing the number of parameters.
3. A **sample-specific gated fusion** of the two modalities, together with a controlled comparison against simple concatenation.
4. A **systematic evaluation** of performance, model size, computational cost, and the role of each component, using a held-out set, 5-fold cross-validation, and ablation experiments.

2 Related Work

2.1 MRI-Based ADHD Classification

In early MRI-based studies of ADHD, handcrafted features were used with conventional classifiers [1]–[3]. For example, Lohani and Rana [5] combined gray-matter volume and cortical-thickness features with personal characteristics and reported 75% accuracy with repeated ten-fold crossvalidation. Later work learns features directly from the data. CNNs have been applied to fMRI [6], [11], graph-based models of resting-state networks have been combined with ensemble classifiers or attention [12], [13], and Pedrollo et al. [9] used CNN ensembles on resting-state fMRI. These studies show that ADHD-related features can be learned from neuroimaging data. The focus, however, is on accuracy: parameter count, FLOPs, memory use, and latency are rarely reported, so it is hard to judge whether such 3D models can run with limited computing resources.

2.2 Multimodal ADHD Classification

Phenotypic features such as age, sex, medication status, and clinical ratings can carry information that the images do not. Deepika et al. [14] combined resting-state functional connectivity with phenotypic data and reported that adding the phenotypic data raised accuracy to 77.59%. A recent preprint has also combined 3D structural MRI with clinical phenotypes and reported 87% accuracy on the Peking-1 subset of ADHD-200. These studies suggest that phenotypic features can complement imaging, but the contribution of each modality is usually not separated within the same model. A gain attributed to multimodal learning may therefore come from the predictive strength of the phenotypic features rather than from extra information in the images.

2.3 Lightweight Models

Depthwise-separable convolution [15], [16] splits spatial filtering and channel mixing into two steps and is a standard way to reduce the number of parameters in a CNN. Efficient channel attention (ECA) [17] reweights feature channels using very few parameters. In ADHD classification, compact models have so far worked on derived features rather than 3D images [10]. To the best of the author’s knowledge, these techniques have not been combined in a small 3D multimodal model for ADHD with the computational cost measured explicitly.

2.4 Adaptive Multimodal Fusion

Concatenation combines the features of each modality in the same way for every sample. Gated multimodal units [18] instead learn gates that weight each modality per sample. No controlled test of whether such gating helps over concatenation for MRI and phenotypic data in ADHD appears to have been reported. This is therefore treated as an open question rather than assumed.

2.5 Research Gap

Existing ADHD studies (Table 1) show the value of neuroimaging, phenotypic features, and deep learning. However, the computational cost and the contribution of each modality and model component are rarely reported. Three questions remain open: (i) can structural MRI and phenotypic features be modeled together with a much smaller 3D model; (ii) can a lightweight 3D model maintain performance similar to a conventional 3D model; and (iii) which model components and modalities actually contribute to the performance? ADHDNet addresses questions (i) and (ii) through its compact design (contributions 1–2), and question (iii) through controlled comparisons and ablation (contributions 3–4).

Table 1. Selected recent ADHD classification studies on ADHD-200 data. Accuracies are not directly comparable because modalities and protocols differ.

Study	Year	Modality	Model	Params	Acc. (%)	Limitations
Lohani & Rana [5]	2023	sMRI + personal char.	SVM/RF/KNN	NR	75.0	Handcrafted features
Gülhan & Özmen [11]	2024	fMRI	3D CNN	NR	76.19 [†]	fMRI only; site-dependent accuracy
Deepika et al. [14]	2024	rs-fMRI + phenotypic	Multimodality model	NR	77.59	Atlas-dependent; no efficiency analysis
Zamanzadeh et al. [12]	2024	rs-fMRI	EasyEnsemble	NR	74.30	Handcrafted graph features
Wu et al. [13]	2025	rs-fMRI	HAGCN	NR	77.95	Requires network construction
Deepika et al. [10]	2025	MRI-derived features	Lightweight attention-KAN	Few thousand	79.25	Derived features, not volumes
ADHDNet (ours)	2026	3D sMRI + phenotype	DS-Conv3D + ECA + gated fusion	0.0716M	90.0[‡]	Phenotype-driven; no external test

NR: not reported. [†]Best per-site accuracy (NeuroIMAGE 76.19%, NYU 69.92%, PU 70.77%).[‡]Held-out split; 5-fold cross-validation: $87.52 \pm 4.61\%$.

3 Materials and Methods

3.1 Dataset

The preprocessed anatomical data of the public ADHD-200 dataset [19], [20] were used. The dataset contains 575 successfully processed subjects (222 ADHD, 353 controls) from six sites. The Pittsburgh site had 78 controls and no ADHD subjects. Because it does not include both classes, it was excluded. The final cohort therefore has 497 subjects (222 ADHD, 275 controls) from five sites (Table 2).

Table 2. ADHD-200 anatomical data by site; Pittsburgh was excluded (no ADHD subjects).

	KKI	NYU	OHSU	Peking_1	NeuroIMAGE	Pittsburgh	Final
ADHD	22	115	36	24	25	0	222
Control	60	96	35	61	23	78	275
Total	82	211	71	85	48	78	497
ADHD (%)	26.8	54.5	50.7	28.2	52.1	0.0	44.7
Status	Incl.	Incl.	Incl.	Incl.	Incl.	Excl.	

3.2 Preprocessing and Phenotypic Features

Each NIfTI volume was resized to $64 \times 64 \times 64$ voxels with trilinear interpolation and min-max normalized to $[0, 1]$. To keep the pipeline simple and reproducible,

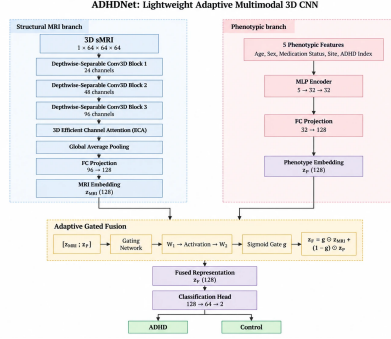


Fig. 1. Architecture of ADHDNet: a depthwise-separable 3D MRI encoder with 3D ECA, an MLP phenotypic encoder, and adaptive gated fusion followed by a classification head.

no additional skull stripping, registration, bias correction, or segmentation was applied. Five phenotypic features were used: age, sex, medication status, acquisition site, and ADHD Index. The ADHD Index is a clinical symptom score that is closely related to the diagnostic label; it is treated as a phenotypic feature, not as an imaging feature. Missing values (coded -999) were filled with the training-set median (continuous) or mode (categorical). Age and ADHD Index were z-scored with training-set statistics, and categorical features were scaled to $[0, 1]$.

4 Proposed ADHDNet

ADHDNet (Fig. 1; 71,552 parameters) has a 3D MRI encoder, a phenotypic encoder, a gated fusion module, and a classification head.

Lightweight 3D MRI Encoder. In 3D CNNs, most parameters and computation come from the convolutions, because a standard 3D convolution with kernel size K needs $K^3 C_{in} C_{out}$ weights. Let $X \in \mathbb{R}^{1 \times 64 \times 64 \times 64}$ be the normalized sMRI volume. The encoder passes X through three blocks that reduce the resolution and increase the number of channels ($24 \rightarrow 48 \rightarrow 96$ channels; $64^3 \rightarrow 8^3$). Each block uses a $3 \times 3 \times 3$ depthwise convolution followed by a $1 \times 1 \times 1$ pointwise convolution [15], [16], then batch normalization [21], ReLU, and $2 \times 2 \times 2$ max pooling:

$$H_l = \text{MaxPool}(\text{ReLU}(\text{BN}(\text{PW}_l(\text{DW}_l(H_{l-1}))))), \quad H_0 = X. \quad (1)$$

Each block then needs only $K^3 C_{in} + C_{in} C_{out}$ weights. This design keeps the 3D structure of the MRI while reducing the number of parameters. Its effect is evaluated in ablation A4.

3D Efficient Channel Attention. Not all feature channels are equally useful, but standard attention modules add fully connected layers that are costly in a small

model. To test whether a very light form of channel attention helps, a 3D version of ECA [17] is applied to the encoder output $F = H_3 \in \mathbb{R}^{96 \times 8^3}$:

$$F_{att} = \sigma(\text{Conv1D}_k(\text{GAP}(F))) \odot F, \quad (2)$$

where GAP is global average pooling. The 1D convolution across channels has a kernel size chosen from the number of channels ($k = 3$ for 96 channels), so the module adds only three parameters. F_{att} is then average-pooled and projected to the MRI embedding $z_{MRI} \in \mathbb{R}^{128}$. Its contribution is evaluated by removing ECA in ablation A1.

Phenotypic Encoder and Gated Fusion. The two modalities may not contribute equally for every subject. A learnable gating function is therefore used to adjust the contribution of each modality per subject. A $5 \rightarrow 32 \rightarrow 32$ MLP with batch normalization, ReLU, and dropout [22] maps the phenotypic features to $h_P \in \mathbb{R}^{32}$. Following gated multimodal units [18], a gating network computes a gate vector g from the 160-dimensional concatenation $[z_{MRI}; h_P]$, and h_P is projected to the phenotype embedding $z_P \in \mathbb{R}^{128}$:

$$g = \sigma(W_2 \delta(W_1[z_{MRI}; h_P])) \in \mathbb{R}^{128}, \quad z_F = g \odot z_{MRI} + (1 - g) \odot z_P, \quad (3)$$

where δ is the ReLU. A $128 \rightarrow 64 \rightarrow 2$ head with dropout maps z_F to class probabilities. Ablation A3 replaces the gate with concatenation, and A2 removes the phenotypic branch.

5 Experimental Setup

Training. All models were implemented in PyTorch [23] and trained with AdamW [24] (learning rate 10^{-4} , weight decay 10^{-4}), batch size 8, at most 150 epochs with early stopping (patience 15), class-weighted cross-entropy with label smoothing [25] of 0.1, gradient clipping at 1.0, and cosine annealing [26]. Training volumes were randomly flipped along each axis ($p = 0.5$). The checkpoint with the best validation AUC was kept, and all seeds were fixed to 42.

Comparator. As a reference, a conventional Conv3D multimodal model was built in the same code base. It uses the same preprocessing, data splits, fusion mechanism, training settings, and evaluation, so differences in cost and performance come mainly from its larger 3D backbone and embeddings. It has four standard Conv3D blocks ($32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ channels) with 3D ECA after blocks 2–4, generalized-mean pooling, a 256-dimensional embedding, a $5 \rightarrow 32 \rightarrow 64 \rightarrow 64$ phenotypic MLP, the same gated fusion, and a $256 \rightarrow 128 \rightarrow 64 \rightarrow 2$ head (1,376,080 parameters).

Evaluation Protocol. (i) *Held-out validation:* a stratified 80:20 split gave 397 training and 100 validation subjects (45 ADHD, 55 controls). Because this validation set was also used for early stopping and model selection, these results are

Table 3. Held-out validation performance of ADHDNet ($n = 100$; ADHD positive).

Accuracy	AUC	Sensitivity	Specificity	Precision	NPV	F1-score
90.00%	0.9543	95.56%	85.45%	84.31%	95.92%	0.8958

**Fig. 2.** Held-out validation results ($n = 100$): (a) ROC curve (AUC = 0.9543); (b) confusion matrix at threshold 0.5.

held-out validation results, not independent test results. (ii) *Cross-validation*: to check whether the held-out result is representative, both models were evaluated with stratified 5-fold cross-validation on all 497 subjects (at most 60 epochs, patience 10; metrics at the best-AUC epoch). (iii) *Ablation*: to measure the effect of depthwise-separable convolution, ECA, the phenotypic branch, and gated fusion, each variant changes one component and keeps everything else the same, including the data split. Imputation and normalization statistics were always computed on the training subjects only, and ADHD was the positive class. FLOPs were counted with PyTorch’s `FlopCounterMode` for one 64^3 volume, and latency was averaged over 50 runs after 10 warm-up runs on a CUDA GPU.

6 Results

6.1 Held-Out Classification Performance

On the held-out validation set (Table 3), an accuracy of 90.00% and an AUC of 0.9543 were obtained by ADHDNet. The best validation AUC occurred at epoch 60, and training stopped at epoch 75. In total, 43 of 45 ADHD subjects and 47 of 55 controls were correctly classified (Fig. 2), with 8 false positives and 2 false negatives. Sensitivity was 95.56%, specificity 85.45%, precision 84.31%, and NPV 95.92%.

6.2 Cross-Validation and Computational Cost

Table 4 compares ADHDNet with the Conv3D comparator. In 5-fold cross-validation, ADHDNet obtained $87.52 \pm 4.61\%$ accuracy (per-fold range 79.80–90.91%) and an AUC of 0.9512 ± 0.0169 , compared with $86.51 \pm 3.64\%$ and

Table 4. ADHDNet versus the Conv3D comparator: complexity per 64^3 volume (latency at batch 1, throughput at batch 8) and 5-fold CV (mean $\pm$ SD).

	Measure	Conv3D comparator	ADHDNet
Complexity	Parameters	1,376,080	71,552
	FP32 weights	5.25 MiB	0.27 MiB
	GFLOPs	6.7952	0.1932
	Latency (ms)	4.32 ± 0.12	2.36 ± 0.15
	Throughput (vol/s)	242.6	662.9
5-fold CV	Accuracy (%)	86.51 ± 3.64	87.52 ± 4.61
	AUC	0.9566 ± 0.0197	0.9512 ± 0.0169
	Sensitivity (%)	85.10 ± 9.63	87.36 ± 5.75
	Specificity (%)	87.64 ± 7.09	87.64 ± 4.71
	F1-score (%)	84.79 ± 4.70	86.21 ± 5.20

Table 5. Ablation results on the held-out validation set ($n = 100$).

Variant	ECA	Pheno.	Fusion	Params	Acc. (%)	AUC	F1
A0 Full ADHDNet	✓	✓	Gated	71,552	90.00	0.9543	0.8958
A1 w/o ECA	–	✓	Gated	71,549	87.00	0.9487	0.8632
A2 w/o phenotype	✓	–	–	28,896	54.00	0.5644	0.4773
A3 Concat. fusion	✓	✓	Concat.	66,272	89.00	0.9564	0.8842
A4 Standard Conv3D	✓	✓	Gated	220,133	89.00	0.9547	0.8842

0.9566 ± 0.0197 for the comparator (Fig. 3a). The two models performed similarly: ADHDNet had slightly higher mean accuracy, sensitivity, and F1-score, but a slightly lower mean AUC. The held-out accuracy of 90.00% is at the upper end of the cross-validation range, so 87.52% is the more representative estimate. With ADHDNet, the number of parameters was reduced by 94.8% ($19.2\times$) and the FLOPs by 97.2% ($35.2\times$), and the FP32 weight storage was reduced from 5.25 to 0.27 MiB (Fig. 3b). Latency was 1.8 times lower and batched throughput 2.7 times higher. A much smaller 3D backbone therefore did not reduce multi-modal performance here. As the ablation shows, however, this performance does not come mainly from the MRI.

6.3 Ablation Study

The ablation variants (Table 5, Fig. 3c) remove ECA (A1), remove the phenotypic branch (A2), replace gated fusion with concatenation (A3), or use standard convolutions (A4). Removing the phenotypic branch reduced performance to near chance (AUC 0.5644). The other variants changed accuracy by at most 3 points. With 100 validation subjects, one subject equals one percentage point, so small differences should be read with caution.

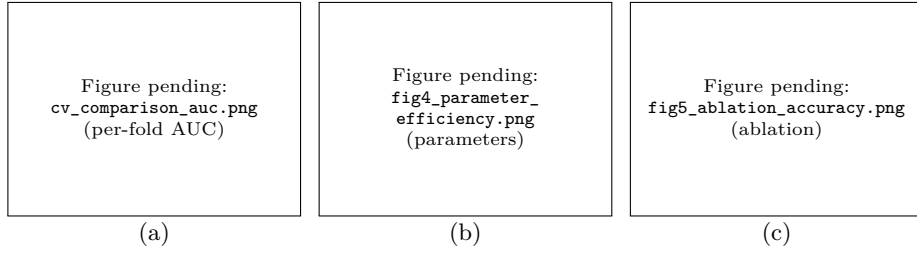


Fig. 3. (a) Per-fold CV AUC; (b) parameters of ADHDNet and the comparator; (c) ablation accuracy.

7 Discussion

Cross-validated performance similar to that of the much larger comparator was obtained with ADHDNet. The ablation, however, shows that this performance comes mainly from the phenotypic features rather than from structural MRI.

Parameter Efficiency. Using standard instead of depthwise-separable convolutions (A4) tripled the number of parameters but did not improve accuracy or AUC on this split. The extra convolutional capacity was therefore not useful in this setting.

Channel Attention. Removing ECA (A1) lowered accuracy by 3.0 points, which is three validation subjects. ECA is best seen as a cheap refinement whose benefit still needs to be confirmed with repeated runs.

Multimodal Fusion. Without the phenotypic branch (A2), performance was close to chance, while every variant that kept it reached an AUC above 0.94. Gated fusion did not clearly outperform concatenation (A3), so its benefit is not yet established.

What the Results Do Not Show. The phenotypic features include the ADHD Index, a symptom score closely related to the diagnosis, and site, whose class proportions differ strongly (Table 2). The reported accuracy therefore does not show that structural MRI, as used here, can distinguish ADHD from controls. It should also not be read as clinically usable diagnostic performance.

Comparison with Prior Work. ADHDNet’s accuracy is numerically higher than the values in Table 1, but those studies use different modalities and protocols. The main point of this study is therefore not higher accuracy. It is the combination of a compact 3D multimodal model with a controlled analysis of its cost and of what each component contributes.

Limitations and Future Work. Performance depends mainly on the phenotypic features. Metrics were taken at the epoch with the best validation AUC, and

there was no external test set. Generalization across sites was not evaluated. The ablation used a single split without repeated runs. The simple preprocessing may limit the anatomical information available to the MRI branch. Finally, only one compact model was evaluated, so the trade-off between cost and performance may differ for other lightweight 3D models.

8 Conclusion

This study investigated whether structural MRI and phenotypic features can be modeled together with a much smaller 3D model. Cross-validated performance similar to that of a conventional Conv3D multimodal model was obtained with ADHDNet, while 94.8% fewer parameters and 97.2% fewer FLOPs were required. The ablation showed that depthwise-separable convolution gave most of the reduction in model size, ECA gave only a small gain, and gated fusion did not clearly outperform concatenation. Most importantly, removing the phenotypic branch reduced performance to near chance, so the reported performance is driven mainly by the phenotypic features. The additional contribution of structural MRI therefore remains unclear. Future work should remove features that are close to the label, such as the ADHD Index and site, and test generalization across sites and on external cohorts.

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